Lab Task 10

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Subject	**SCD**

Refactored Code

```
package project2;

import java.util.Scanner;

abstract class Operation {
    protected double num1;
    protected double num2;

    public Operation(double num1, double num2) {
        this.num1 = num1;
        this.num2 = num2;
    }

    public abstract double calculate();
    }


class Addition extends Operation {
    public Addition(double num1, double num2) {
        super(num1, num2);
    }

    public double calculate() {
        return num1 + num2;
    }
}

    class Subtraction extends Operation {
    public Subtraction(double num1, double num2) {
        super(num1, num2);
    }


    public double calculate() {
        return num1 - num2;
    }
}

class Multiplication extends Operation {
    public Multiplication(double num1, double num2) {
        super(num1, num2);
    }


    public double calculate() {
        return num1 * num2;
    }
}

class Division extends Operation {
    public Division(double num1, double num2) {
        super(num1, num2);
    }


    public double calculate() {
        if (num2 == 0) {
            throw new ArithmeticException("Cannot divide by zero");
        }
        return num1 / num2;
    }
}
```

```java

public class Calculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);


        System.out.println("Simple Calculator");
        double num1 = getNumberInput(scanner, "Enter the first number: ");
        double num2 = getNumberInput(scanner, "Enter the second number: ");
        char operator = getOperatorInput(scanner);


        try {
            Operation operation = createOperation(num1, num2, operator);
            double result = operation.calculate();
            System.out.println("Result: " + result);

        } catch (IllegalArgumentException | ArithmeticException e) {

            System.out.println("Error: " + e.getMessage());

        } finally {

            scanner.close();
        }
    }


    private static double getNumberInput(Scanner scanner, String prompt) {
        System.out.print(prompt);
        return scanner.nextDouble();
    }


    private static char getOperatorInput(Scanner scanner) {
        System.out.print("Enter the operation (+, -, *, /): ");
        return scanner.next().charAt(0);
    }


    private static Operation createOperation(double num1, double num2, char operator) {
        switch (operator) {
            case '+': return new Addition(num1, num2);
            case '-': return new Subtraction(num1, num2);
            case '*': return new Multiplication(num1, num2);
            case '/': return new Division(num1, num2);
            default: throw new IllegalArgumentException("Invalid operator. Please enter +, -, *, or /");
        }
    }
}

```

Output

```
Simple Calculator
Enter the first number: 10
Enter the second number: 20
Enter the operation (+, -, *, /): +
Result: 30.0
```